Lesson notes · for you

Exploring a dataset

Data science · Level 2 · Lesson 4

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Follow a five-question routine.
- Spot gaps and oddities.
- Write findings with evidence.

Activity: Five questions, three findings

1. Warm-up (5 min): Look at a big table and say the first thing you would check.
2. Explore (10 min): With a partner, work through the five questions for a small table.
3. Write (10 min): Write three findings with numbers and sources, and one next question.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Which question found a surprise?
- What makes a finding good?
- What is your next question?

Check yourself

- Write one finding from today with a number and a source.

Try it this week
Pick a small table, such as a week of weather or your class's library loans, and go through the five questions. Write three findings with numbers and one next question.

Worksheet

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Name _____ Class _____ Date _____

1 **Which is the better finding?**

- ◯ Attendance is bad
- ◯ Attendance falls to a median of 80% in July
- ◯ Attendance changes during the year

2 **Put the five questions in order.**

- ☐ Relationships
- ☐ Gaps and oddities
- ☐ Centre and spread
- ☐ What shape?
- ☐ What is in it?

3 **Write summary(xs) to return the smallest value, the median and the largest. xs has an odd length.**

```
def summary(xs):
    return 0, 0, 0

print(summary([7, 9, 6, 25, 8]))
```

Write your program:

4 **Your screen-time column is skewed right. What should you report?**

- ◯ The median and the range
- ◯ Only the mean
- ◯ Only the largest value

5 **Which are good findings? Choose all that apply.**

- ☐ On school nights my median sleep is 7 hours.
- ☐ I'm always tired.
- ☐ Two days have no screen time recorded.
- ☐ Sleep is important.

✎ **What would you like to explore about your own week, if you kept the data?**